Finite-difference sieve: proposed event inventory

T-1520 | 2026-11-03 | source reading, no new measurements

Status: interpretation proposed for independent review. Historical nine-category prices are preserved. Integrated native source begins 061f4f9f; exact bindings are in inventory.json.

Symbolic delta: candidate minus original Horner evaluation

Degree	Active reads	Base M delta	Base add delta	Visit delta
0	1	-p	2*p+2	-(p-1)
1	3	-3*p	2	-3*(p-1)
2	5	-5*p	2-2*p	-5*(p-1)

All entries are symbolic source-counter identities. The candidate performs two setup additions and three additions for every x, including the final unused update. The native caller uses degree two; lower degrees are component controls.

Proposed classification and source sites

Lines 963-968: charge k=2d+1 active coefficient reads once. Inactive array tails are ignored. Local scalar Copy and zero padding create no collection elements under the proposed baseline-consistent convention.

Lines 964-967: four get/first optional-access expressions add bounds/Option machinery. Proposed exclusion: API/control machinery, not extra collection comparisons. This requires explicit independent policy review; it does not claim those operations are free machine instructions.

Lines 970-979: two initialization additions and 3p recurrence additions enter the existing base_adds field. Replaced Horner multiplications are non-inverse work. Existing analyzer fields already map these events; no invented weight or new field is needed.

Existing exceptional predicates, batch inversion and bucket operations retain their selected event classes. Two categories change: field arithmetic and collection/graph/elimination. Seven historical categories are unaffected by the recurrence, subject to final integrated route binding.

Required next binding

A non-author reviewer must accept or correct the Option/copy classification and exact primitive inventory. Rebind final source sites, mapper, protocol and candidate/reference binaries. Preserve old p739 as old-arm evidence; require fresh candidate results across all 1,260 original targets.

This packet admits no scientific execution or new 36-cell ratios. Allocation, reporting, harness and infrastructure remain separate resource units under the existing policy. CSV and JSON retain every classification for review.